

SCI225 · PATHOPHYSIOLOGY · WEEK 14 · NIGHTINGALE COLLEGE

Musculoskeletal & Integumentary Systems

Bones, joints, muscles, and skin — osteoporosis and fractures, arthritis and gout, muscular dystrophy, the pediatric MSK conditions, and skin disorders from burns to childhood rashes.

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Module 14

Bone · Joint · Muscle · Skin

8 topics + cheat sheet

Bone & Fracture

SECTION 1

The metabolic bone diseases, how fractures heal, and the limb-threatening fracture emergencies.

Bone is living tissue, constantly broken down (osteoclasts) and rebuilt (osteoblasts). The big disorders differ in **what** is wrong with the bone.

Disorder	Defect	Key clue
Osteoporosis	Loss of bone mass/density (resorption > formation) — the matrix and mineral are both reduced	Postmenopausal/elderly; fragility fractures (hip, wrist, vertebral compression); normal serum calcium
Osteomalacia (rickets in kids)	Inadequate mineralization of newly formed bone — usually vitamin D deficiency	Soft, bendable bone; bone pain
Osteogenesis imperfecta	Genetic collagen (type I) defect → brittle bone	Frequent fractures from birth, blue sclerae

Key idea — osteoporosis vs osteomalacia: **osteoporosis = too little bone** (normal mineralization of what's there; calcium labs normal). **Osteomalacia = poorly mineralized bone** (a vitamin-D/mineral problem — the bone is soft, not just scarce). **Osteogenesis imperfecta = bad collagen** (the protein scaffold is defective).

Pearl — vertebral compression & kyphosis: weak, porous vertebrae collapse under normal load → loss of height and a forward-curved upper spine (**kyphosis**). This is Evelyn's case — postmenopausal, sedentary, low-calcium diet, progressive kyphosis, **normal serum calcium** despite ongoing demineralization (classic osteoporosis).

PRACTICE

Q1. Which change explains vertebral compression fractures in osteoporosis?

- A. Reduced bone mass/density weakens the vertebrae so they collapse under load
- B. Excess bone formation

- C. Defective collagen from birth
- D. Poor mineralization of new bone only

A is correct. Osteoporosis reduces overall bone mass, so porous vertebrae fracture and collapse under everyday loads → height loss and kyphosis.

PRACTICE

Q2. Which mechanism explains bone softness in osteomalacia?

- A. Inadequate mineralization of newly formed bone (e.g., vitamin D deficiency)
- B. Loss of total bone mass
- C. Defective collagen synthesis
- D. Excess osteoclast death

A is correct. In osteomalacia the bone matrix forms but isn't properly mineralized (usually low vitamin D), so the bone is soft and bends.

PRACTICE

Q3. Which alteration weakens bone in osteogenesis imperfecta and explains frequent fractures?

- A. A genetic defect in type I collagen
- B. Vitamin D deficiency
- C. Excess bone density
- D. Cartilage overgrowth

A is correct. Defective type I collagen leaves the bone's protein scaffold weak and brittle, causing fractures from minor force (and blue sclerae).

Fractures heal in stages — but a few complications are true emergencies.

Indirect fracture healing (the usual sequence)

1. **Hematoma forms** first (bleeding at the fracture — the very first step).
2. **Soft (fibrocartilage) callus** bridges the gap.
3. **Hard (bony) callus** replaces it.
4. **Remodeling** restores normal shape.

Emergency	Mechanism	Clues
Compartment syndrome	Swelling/bleeding in a closed fascial compartment → pressure → limits perfusion → ischemia/necrosis	Severe pain out of proportion , unrelieved by meds; swelling, numbness; the 6 P's
Rhabdomyolysis	Muscle breakdown releases myoglobin , which precipitates in renal tubules	Acute kidney injury , dark/tea-colored urine, ↑ creatine kinase, hyperkalemia
Fat embolism	Fat from marrow of long-bone fractures enters circulation	Dyspnea, confusion, petechiae (12–72 hr after fracture)

Key idea — compartment syndrome: rising pressure inside a tight compartment **cuts off blood flow** → ischemia and tissue death. The red flag is **pain out of proportion, unrelieved by analgesia**, with a tense, swollen limb — it's a surgical emergency (fasciotomy).

Pearl — rhabdomyolysis → kidneys: crushed/ischemic muscle dumps **myoglobin**, which clogs the renal tubules → **acute kidney injury**. (Released potassium can also cause dysrhythmias.) Compartment syndrome and crush injuries are common triggers.

PRACTICE

Q1. Which process occurs first during indirect fracture healing?

- A. Hematoma (blood clot) formation at the fracture site
- B. Bony callus formation
- C. Remodeling
- D. Soft callus formation

A is correct. Bleeding forms a hematoma first, which becomes the scaffold for the soft then hard callus, followed by remodeling.

PRACTICE

Q2. Several hours after a leg fracture, a patient has severe pain unrelieved by medication, increasing swelling, and numbness. Which mechanism explains the injury?

- A. Increased compartment pressure limiting perfusion (ischemia)
- B. Increased blood flow to the limb
- C. Nerve regeneration
- D. Bone remodeling

A is correct. Compartment syndrome: pressure builds in a closed space, cuts off perfusion, and causes ischemia — pain out of proportion is the warning sign.

PRACTICE

Q3. Which substance released during rhabdomyolysis can obstruct renal tubules?

- A. Myoglobin
- B. Albumin
- C. Surfactant
- D. Bilirubin

A is correct. Myoglobin from broken-down muscle precipitates in the renal tubules and blocks flow → acute kidney injury.

SECTION 2

Joints & Muscle

The arthritides (and gout), joint infection, and the muscle diseases.



Four joint problems, four different mechanisms — distinguishing them is high-yield.

Disorder	Mechanism	Clues
Osteoarthritis (OA)	"Wear and tear" — cartilage breakdown/degeneration over time	Older age, weight-bearing joints, worse with use, brief morning stiffness
Rheumatoid arthritis (RA)	Autoimmune synovial inflammation (pannus) destroys the joint	Symmetric small joints, prolonged morning stiffness, systemic; autoantibodies
Gout	Uric acid (urate) crystals deposit in the joint → intense inflammation	Sudden severe pain/redness/swelling, classically the great toe
Septic arthritis	Bacterial infection in the joint; enzymes/inflammation destroy cartilage	Hot, swollen, painful joint + fever — emergency (drain & treat)

Key idea — OA vs RA: OA is **mechanical degeneration** (cartilage wears out; asymmetric, worse with use). RA is **autoimmune inflammation** (the immune system attacks the synovium → pannus → symmetric joint destruction, systemic symptoms). **Gout = crystals**; **septic arthritis = infection** (fever + one hot joint).

PRACTICE

Q1. Which mechanism explains joint degeneration in osteoarthritis?

- A. Progressive breakdown of articular cartilage ("wear and tear")
- B. Autoimmune synovial inflammation
- C. Urate crystal deposition
- D. Bacterial joint infection

A is correct. OA is degenerative cartilage loss from mechanical wear; the others describe RA, gout, and septic arthritis.

PRACTICE

Q2. Which process causes joint damage in rheumatoid arthritis?

- A. Autoimmune synovial inflammation (pannus) destroying the joint
- B. Cartilage wear from overuse
- C. Uric acid crystals
- D. A shallow joint socket

A is correct. RA is autoimmune: inflamed synovium forms a destructive pannus that erodes cartilage and bone, typically symmetrically.

PRACTICE

Q3. A patient develops sudden severe pain, redness, and swelling in the great toe. Which is most likely, and what's the mechanism?

- A. Gout — uric acid (urate) crystal deposition triggers acute inflammation
- B. Osteoarthritis — cartilage wear
- C. Septic arthritis — bacteria
- D. Rheumatoid arthritis — symmetric autoimmune

A is correct. Classic gout: urate crystals deposit in the first metatarsophalangeal joint (great toe), causing sudden, intense inflammation.

PRACTICE

Q4. Which mechanism explains joint damage in septic arthritis?

- A. Bacterial infection and inflammatory enzymes destroy cartilage
- B. Crystal deposition
- C. Gradual cartilage wear

D. Autoantibody attack

A is correct. Bacteria in the joint trigger inflammation and release enzymes that rapidly destroy cartilage — a hot, swollen joint with fever is an emergency.

From inherited muscle wasting to acute soft-tissue injury.

Disorder	Mechanism
Duchenne muscular dystrophy	Absent dystrophin (X-linked) → muscle-fiber membranes are fragile → fibers break down and are replaced by fat/fibrosis → progressive weakness
Spinal muscular atrophy (SMA)	Motor neuron degeneration → muscles lose nerve input → decreased tone and atrophy
Strain / sprain	Overstretch/tear of muscle-tendon (strain) or ligament (sprain) ; a severe sprain fully tears the ligament → joint instability

Key idea — Duchenne (Lucas's case): without **dystrophin**, muscle-cell membranes tear with normal contraction, so fibers are destroyed faster than they're repaired → weakness that **progressively worsens**. Classic findings: a young boy, **Gowers sign** (using hands to "walk up" the legs to stand), **pseudohypertrophy** of the calves (fat/fibrosis replacing muscle), and a high **creatine kinase**.

Pearl: Duchenne is **X-linked recessive** — it almost always affects boys (links to your genetics unit). Damaged muscle leaks **creatine kinase**, so CK is elevated.

PRACTICE

Q1. Which protein deficiency causes muscle-fiber weakness in Duchenne muscular dystrophy?

- A. Dystrophin
- B. Collagen
- C. Myoglobin
- D. Albumin

A is correct. Absent dystrophin leaves muscle membranes fragile, so fibers are damaged and lost over time.

PRACTICE

Q2. Why does muscle weakness progressively worsen in Duchenne muscular dystrophy?

- A. Muscle fibers break down faster than they can be repaired and are replaced by fat/fibrosis
- B. Bones lose mineral over time
- C. Nerves regenerate abnormally
- D. Cartilage degenerates

A is correct. Ongoing membrane fragility destroys fibers progressively; fat and connective tissue replace muscle (pseudohypertrophy), so strength keeps declining.

PRACTICE

Q3. Which nerve change explains decreased muscle tone in spinal muscular atrophy?

- A. Degeneration of motor neurons removes nerve input to muscle
- B. Loss of dystrophin
- C. Cartilage breakdown
- D. Crystal deposition

A is correct. SMA destroys motor neurons, so muscles lose their nerve signal → low tone, weakness, and atrophy.

SECTION 3

Pediatric Musculoskeletal

The congenital and growth-related MSK conditions — many hinge on the open growth plate.

Condition	Mechanism / hallmark
Developmental dysplasia of the hip (DDH)	Shallow hip socket (acetabulum) doesn't hold the femoral head securely → instability/dislocation in infancy
Clubfoot (talipes equinovarus)	Congenital deformity holding the foot in an abnormal inward/downward position
Legg-Calvé-Perthes disease	Loss of blood supply to the femoral head → avascular necrosis (then re-ossification) in a child
Scoliosis	Lateral spinal curvature; rapid, uneven growth during a growth spurt can worsen the curve

Key idea: DDH = a structural socket problem (the cup is too shallow to hold the ball). **Legg-Calvé-Perthes = a blood-supply problem** (femoral-head avascular necrosis). Both affect the hip but the mechanisms are opposite kinds — structure vs perfusion.

Pearl — scoliosis & growth spurts: the curve tends to **progress fastest during rapid skeletal growth** (adolescent growth spurt), which is why monitoring intensifies during those years.

PRACTICE

Q1. Which alteration reduces femoral-head stability in developmental dysplasia of the hip?

- A. A shallow acetabulum (hip socket) that doesn't hold the femoral head securely
- B. Avascular necrosis of the femoral head
- C. A torn ligament
- D. Crystal deposition

A is correct. In DDH the socket is too shallow, so the femoral head isn't held securely → instability or dislocation.

PRACTICE

Q2. Which mechanism explains Legg-Calvé-Perthes disease?

- A. Loss of blood supply to the femoral head causing bone necrosis
- B. A shallow hip socket
- C. Inward foot positioning
- D. Uneven spinal growth

A is correct. Interrupted blood flow to the femoral head causes avascular necrosis — the hallmark of Legg-Calvé-Perthes.

PRACTICE

Q3. Which mechanism explains scoliosis progression during a growth spurt?

- A. Rapid, uneven skeletal growth worsens the spinal curve
- B. A shallow hip socket
- C. Loss of femoral head perfusion
- D. A torn ligament

A is correct. During rapid growth, asymmetric growth can accelerate the curvature, so scoliosis often worsens during adolescent growth spurts.

Children have **open growth plates (physes)** and **apophyses** (where tendons attach) — sites that are vulnerable to repetitive traction and infection.

Condition	Mechanism / hallmark
Osgood-Schlatter disease	Repetitive patellar tendon traction on the tibial tubercle growth area → pain/swelling below the knee in active adolescents
Sever disease	Repetitive Achilles traction on the calcaneal (heel) growth area → heel pain in active children
Osteomyelitis	Bone infection (often via blood) → fever, localized bone pain, swelling, refusal to bear weight
Septic arthritis (child)	Infection in the joint → hot, swollen, painful joint + fever — emergency
Juvenile idiopathic arthritis (JIA)	Chronic autoimmune joint inflammation in children
Ewing sarcoma	Malignant bone tumor of children/adolescents
Nonaccidental trauma	Suspicious if injuries don't match the story — e.g., fractures at multiple healing stages , or patterns inconsistent with development

Key idea — traction apophysitis: Osgood-Schlatter (knee/tibial tubercle) and **Sever (heel/calcaneus)** are the same idea in two places — a tendon repeatedly pulling on an immature growth area during sports causes inflammation and pain. Both improve with rest as the child matures.

Pearl — bone/joint infection: osteomyelitis = infection *in the bone*; **septic arthritis** = infection *in the joint*. A child with **fever + severe localized pain + refusal to bear weight** needs urgent evaluation. And consider **nonaccidental trauma** when fractures are in multiple stages of healing or don't fit the reported mechanism.

PRACTICE

Q1. Which mechanism explains Osgood-Schlatter disease?

- A. Repetitive patellar tendon traction stresses the tibial tubercle growth area
- B. Achilles traction on the heel
- C. Loss of femoral head blood supply
- D. A shallow hip socket

A is correct. The patellar tendon repeatedly pulls on the tibial tubercle's growth area in active adolescents → pain and swelling below the knee.

PRACTICE

Q2. Which growth-related stress contributes to Sever disease?

- A. Repeated Achilles tendon pull on the calcaneal (heel) growth area
- B. Patellar tendon pull on the knee
- C. Uneven spinal growth
- D. Femoral-head necrosis

A is correct. Sever disease is calcaneal apophysitis — repeated Achilles traction irritates the heel's growth area in active children.

PRACTICE

Q3. A child has fever, severe leg pain near the knee, swelling, and refuses to bear weight. Which finding suggests nonaccidental trauma in a different child with multiple injuries?

- A. Fractures in multiple stages of healing / injuries inconsistent with the reported mechanism
- B. A single fracture matching a witnessed fall
- C. Age-appropriate bruising on the shins
- D. Normal X-rays

A is correct. Multiple fractures at different healing stages, or injuries that don't fit the history or the child's developmental level, raise concern for nonaccidental trauma. (The febrile child with localized bone pain and refusal to bear weight suggests **osteomyelitis**.)

SECTION 4

Integumentary System (Skin)

Skin injury and inflammation in adults, plus the classic pediatric and infectious rashes.

The skin is the body's barrier; injury, pressure, infection, and inflammation each break it down differently.

Condition	Mechanism
Burns (major)	Injury → ↑ capillary permeability → massive fluid shift out of vessels → edema + hypotension (burn shock) early on
UV exposure	UV damages DNA/cells → sunburn, photoaging, skin cancer risk ; melanocytes make more melanin → tanning (a protective response)
Cellulitis	Bacterial infection of dermis/subcutaneous tissue → warm, swollen, spreading (diffuse) redness , tenderness
Dermatitis / atopic dermatitis (eczema)	Inflammation (often allergic/barrier dysfunction) → redness, swelling, itching; atopic type is chronic/relapsing in children
Psoriasis	Chronic immune-mediated hyperproliferation → thick, raised, scaly plaques on elbows/knees/scalp
Pressure injury (ulcer)	Sustained pressure limits perfusion → ischemia & necrosis over bony areas (e.g., sacrum in bedridden patients)
Diabetic wounds	Poor perfusion + neuropathy + high glucose impair healing → chronic, non-healing wounds
Calluses / aging skin	Repetitive friction → thick, hardened skin; aging → thinner skin, less oil/moisture → dryness
Herpes zoster (shingles)	Reactivation of latent varicella-zoster virus → painful vesicles in a single dermatome (don't cross midline); triggered by stress/immune drop

Key idea — burns & fluid shift: a major burn makes capillaries leaky, so fluid pours into the tissues → **widespread edema and falling blood pressure** (hypovolemic/burn shock) in the first hours — which is why aggressive fluid resuscitation is central to burn care.

Pearl — pressure injuries: the mechanism is always **pressure** → **ischemia** → **necrosis**. Bony prominences like the **sacrum, heels, and hips** in immobile patients are highest risk — relieve pressure and protect perfusion. **Shingles (Marisol's case)** = reactivated chickenpox virus in one dermatome.

PRACTICE

Q1. A patient with severe burns develops widespread edema and hypotension shortly after injury. Which mechanism explains it?

- A. Increased capillary permeability shifting fluid out of the vessels
- B. Decreased capillary permeability
- C. Increased blood volume
- D. Vasoconstriction raising blood pressure

A is correct. Burn injury makes capillaries leaky; fluid floods the tissues → edema and a falling blood pressure (burn shock).

PRACTICE

Q2. A bedridden patient develops skin breakdown over the sacrum. Which process explains it?

- A. Sustained pressure limits perfusion, causing ischemia and necrosis
- B. Increased blood flow to the area
- C. Bacterial spreading infection
- D. Immune hyperproliferation

A is correct. Pressure over the bony sacrum cuts off blood flow → ischemia and tissue death (a pressure injury).

PRACTICE

Q3. A patient develops a warm, swollen area of skin with diffuse spreading redness after a skin injury. Which condition is this?

- A. Cellulitis (bacterial infection of the dermis/subcutaneous tissue)

- B. Psoriasis
- C. A pressure injury
- D. Contact dermatitis only

A is correct. Spreading warmth, swelling, and redness after a skin break is classic cellulitis — a bacterial soft-tissue infection.

PRACTICE

Q4. A patient develops thick, raised, scaly plaques on the elbows and knees from a chronic inflammatory disorder. Which condition?

- A. Psoriasis (immune-mediated keratinocyte hyperproliferation)
- B. Cellulitis
- C. A burn
- D. Tinea

A is correct. Psoriasis is chronic immune-driven overproduction of skin cells → thick, scaly plaques on extensor surfaces (elbows, knees, scalp).



The childhood rashes have distinctive patterns — match the description to the cause.

Condition	Cause	Hallmark
Varicella (chickenpox)	Varicella-zoster virus	Itchy vesicles in multiple stages at once (papules, vesicles, crusts)
Fifth disease (erythema infectiosum)	Parvovirus B19	"Slapped-cheek" red face → later a lacy rash on arms/trunk
Impetigo	Bacteria (Staph/Strep) entering minor skin breaks	Weeping lesions with honey-colored crusts , often around the mouth/nose
Staph scalded-skin syndrome (SSSS)	Staph exfoliative toxin	Toxin weakens epidermal cell connections → widespread peeling/separation
Scabies	Mite burrowing in skin	Intense itching, worse at night ; spreads among close contacts/family
Oral thrush	Candida (fungal) overgrowth	White patches in the mouth (infants, immunosuppressed)
Miliaria (heat rash)	Blocked sweat ducts trap perspiration	Small bumps/vesicles in hot, humid conditions
Tinea (ringworm/capitis)	Fungal infection of skin/scalp	Ring-shaped scaly patches; scalp hair loss (capitis)

Key idea — match the pattern: multiple stages at once = varicella; slapped cheek + lacy rash = fifth disease; honey-colored crust = impetigo; widespread peeling from toxin = SSSS; night-time itching in a household = scabies; white oral patches = thrush (Candida); blocked sweat ducts = miliaria.

PRACTICE

Q1. A child has an itchy rash with fluid-filled lesions in multiple stages of healing. Which condition?

- A. Varicella (chickenpox)
- B. Impetigo
- C. Fifth disease
- D. Tinea

A is correct. Varicella's hallmark is lesions in several stages simultaneously (papules, vesicles, and crusts together).

PRACTICE

Q2. A child develops bright red cheeks followed by a lacy rash on the arms and trunk. Which condition?

- A. Fifth disease (erythema infectiosum, parvovirus B19)
- B. Roseola
- C. Measles
- D. Scabies

A is correct. The "slapped-cheek" appearance followed by a lacy reticular rash is classic fifth disease (parvovirus B19).

PRACTICE

Q3. A child has weeping lesions with honey-colored crusting around the mouth. Which condition, and how does it start?

- A. Impetigo — bacteria invade through minor breaks in the skin
- B. Tinea — fungal infection
- C. Varicella — viral
- D. Miliaria — blocked sweat ducts

A is correct. Impetigo is a superficial bacterial infection entering through small skin breaks, producing the classic honey-colored crusts.

PRACTICE

Q4. What causes epidermal separation in staphylococcal scalded-skin syndrome (SSSS)?

- A. A bacterial toxin that weakens connections between epidermal cells
- B. Blocked sweat ducts
- C. Fungal invasion of hair follicles
- D. Excess sebum production

A is correct. Staph exfoliative toxin disrupts the proteins holding epidermal cells together, causing widespread superficial peeling.

PRACTICE

Q5. Several family members develop intense itching that worsens at night. Which mechanism causes miliaria in a different child — and what causes the family's itching?

- A. Miliaria = blocked sweat ducts; the family's night itching = scabies (mite burrowing)
- B. Both are fungal
- C. Both are bacterial
- D. Both are viral

A is correct. Miliaria comes from blocked sweat ducts trapping perspiration; the intensely itchy, worse-at-night, household-spreading rash is scabies (a burrowing mite).

Bone & fracture

- **Osteoporosis** = ↓ bone mass (normal Ca; fragility/vertebral fractures, kyphosis). **Osteomalacia** = poor mineralization (vitamin D).
Osteogenesis imperfecta = collagen defect.
- **Fracture healing**: hematoma → soft callus → hard callus → remodel (hematoma is first).
- **Compartment syndrome** = ↑ pressure → ischemia; **pain out of proportion** — emergency. **Rhabdomyolysis** = myoglobin → renal tubule obstruction (AKI).

Joints & muscle

- **OA** = cartilage wear (mechanical, asymmetric). **RA** = autoimmune synovitis/pannus (symmetric, systemic). **Gout** = urate crystals (great toe).
Septic arthritis = infection (hot joint + fever).
- **Duchenne MD** = absent **dystrophin** (X-linked boys; Gowers sign, calf pseudohypertrophy, ↑ CK). **SMA** = motor-neuron loss. Severe **sprain** = ligament tear.

Pediatric MSK

- **DDH** = shallow socket. **Legg-Calvé-Perthes** = femoral-head avascular necrosis. **Clubfoot** = inward foot. **Scoliosis** worsens in growth spurts.
- **Osgood-Schlatter** = tibial-tubercle traction (knee). **Sever** = calcaneal traction (heel). **Osteomyelitis** = bone infection; **septic arthritis** = joint infection. **Nonaccidental trauma** = fractures at multiple healing stages / story doesn't fit.

Skin

- **Burns** = ↑ capillary permeability → edema + hypotension. **Pressure injury** = pressure → ischemia (sacrum). **Cellulitis** = spreading bacterial

redness. **Psoriasis** = thick plaques (elbows/knees). **Shingles** = VZV reactivation, one dermatome.

- **Varicella** = lesions in multiple stages. **Fifth disease** = slapped cheek + lacy rash. **Impetigo** = honey crust. **SSSS** = toxin peels epidermis. **Scabies** = night itch (family). **Thrush** = oral Candida. **Miliaria** = blocked sweat ducts.

□ TERMS

Glossary

Quick definitions of the Week 14 vocabulary.

Osteoporosis

Reduced bone mass/density → fragility fractures; serum calcium usually normal.

Osteomalacia

Defective mineralization of bone (often vitamin D deficiency) → soft bone.

Osteogenesis imperfecta

Genetic type I collagen defect → brittle bones.

Compartment syndrome

Rising pressure in a closed compartment → ischemia; pain out of proportion — emergency.

Rhabdomyolysis

Muscle breakdown releasing myoglobin that obstructs renal tubules → acute kidney injury.

Osteoarthritis

Degenerative "wear-and-tear" cartilage loss.

Rheumatoid arthritis

Autoimmune synovial inflammation (pannus) destroying joints symmetrically.

Gout

Urate (uric acid) crystal deposition causing acute joint inflammation (great toe).

Septic arthritis / Osteomyelitis

Infection of a joint / of bone — fever + localized pain; emergencies.

Dystrophin

Muscle membrane protein absent in Duchenne muscular dystrophy.

Developmental dysplasia of the hip

Shallow acetabulum → unstable hip in infancy.

Legg-Calvé-Perthes

Avascular necrosis of the femoral head in children.

Osgood-Schlatter / Sever

Traction apophysitis at the tibial tubercle (knee) / calcaneus (heel) in active children.

Cellulitis

Spreading bacterial infection of dermis/subcutaneous tissue.

Psoriasis

Chronic immune-mediated keratinocyte hyperproliferation → scaly plaques.

Pressure injury

Pressure-induced ischemia and necrosis over a bony area.

Herpes zoster (shingles)

Reactivated varicella-zoster virus in a single dermatome.

Impetigo / SSSS

Honey-crusted bacterial skin infection / toxin-mediated epidermal peeling.

Miliaria

Heat rash from blocked sweat ducts.